Supplementary materials:
Scenario audit of Cuba's 2021–2026 population decline

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Status. Model E is *retired*. It is documented here only for traceability: it was superseded by the age-standardised excess-mortality decomposition in the main paper, which estimates the same quantity from observed age denominators rather than a hand-weighted dimensionless index. Its figures are historical and are not current claims.

These materials accompany the main preprint. Models A–D are scenarios; Model D is an illustrative central path, not an identified estimator.

1 Provenance of headline quantities

Table 1 classifies major inputs as observed (ONEI tables), derived (accounting identities), assumed (scenario priors), or expert-coded (HSDS features). Full machine-readable version: `data/provenance.yaml`.

Table 1: Provenance taxonomy (selected quantities).

Quantity	Class	Source	Status
ONEI B, D, 2022–25	observed	Anuario / Indicadores	canonical
Identity R	derived	`model_from_2021.py`	canonical
Scenario envelope 1.97–2.79 M	assumed	scenario priors	primary
Central scenario 8.58 M	assumed	central margins	scenario v
Destination floor 1.05 M	derived	`destinations_settled.csv`	author-ass
Excess vs the 2019 schedule (2024–25 $\sim$50k)	derived	`attributable_mortality.py`	age-standa
Crude 2019-CDR residual $\sim$77k	derived	`mortality_schedule_residual.py`	*supersede*
HSDS S_t	expert-coded	`hsds_features.csv`	Model E c

2 Life expectancy series

Official life expectancy at birth peaked in 2011–13 (78.5 years) and fell to 77.7 in 2018–20 (Figure 1). An independent 2021 estimate places it near 71 years [Albizu-Campos Espiñeira, 2023]; Brønnum-Hansen et al. [2023] covers 1955–2020 and does not extend to 2021. Life expectancy re-expresses the same mortality shock rather than providing independent corroboration of Model D; its magnitude is large even relative to COVID-era losses documented internationally [Aburto et al., 2022, Msemburi et al., 2023].

3 Provincial intensity of exit

Subnational net outflow is national but uneven (Figure 2). ONEI *Indicadores 2025* reports Havana external net emigration near $-$**33 per 1,000** in 2025; western and central provinces empty faster than the east [ONEI,

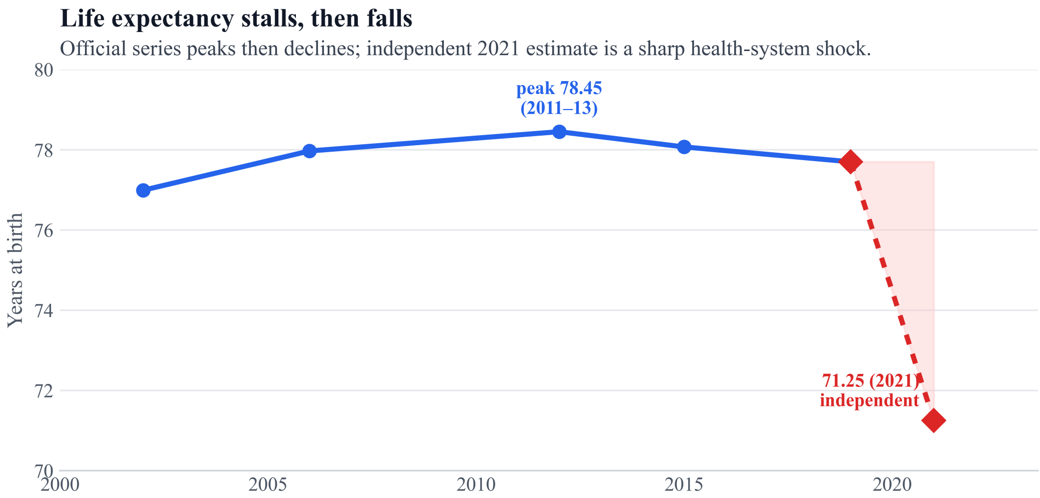

Figure 1: Life expectancy at birth: official triennial series and independent 2021 estimate.

2025]. If national migration accounting is biased, provincial *ranks* are more trustworthy than precise levels.

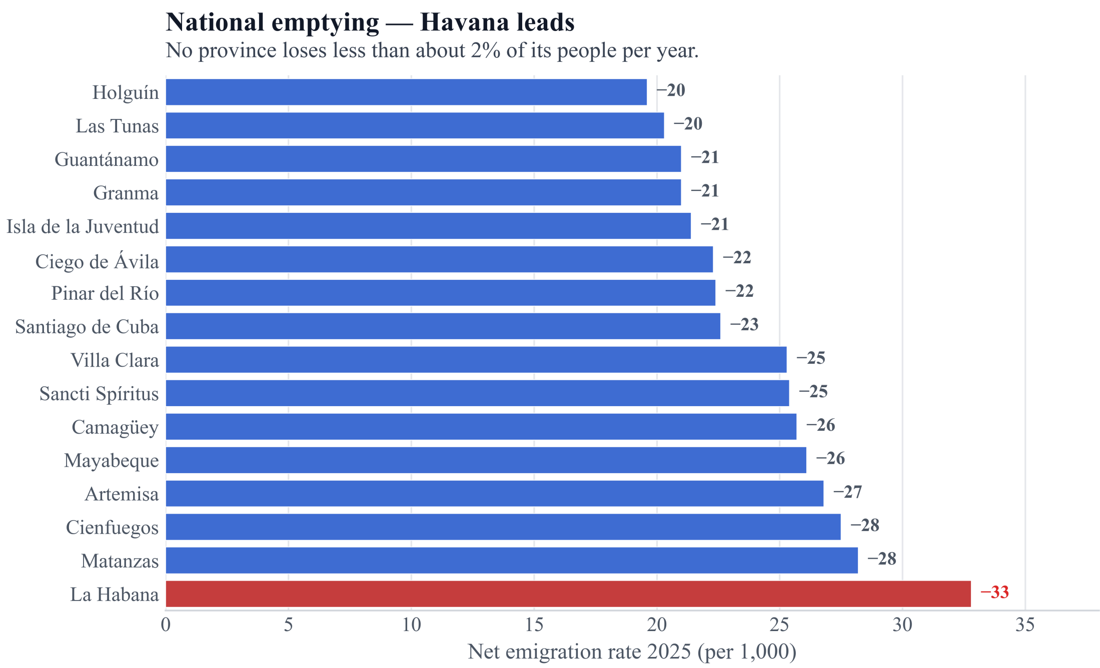

Figure 2: Net emigration rate by province, 2025 (ONEI-reported).

4 Retired models

Two models were built and then retired. They are recorded here for traceability, without their numeric results: a retired model's headline figure that stays in a distributed document will be quoted as if it were current, which is the specific failure this supplement previously had.

Model B was dropped because it landed within 0.02 M of the crisis-adjusted central scenario—the same scenario with different dials.

Model E reconstructed deaths and births from a yearly health-system deterioration score S_t, then closed the identity with a migration bridge, in the spirit of excess-mortality reconstruction under incomplete registration [Karlinsky and Kobak, 2021, Msemburi et al., 2023, Kishore et al., 2018]. It was superseded by the age-standardised decomposition in the main paper, which estimates the same quantity from observed age denominators rather than a hand-weighted dimensionless index. It failed its own adversarial gate on six counts: ageing entered both the expected-death schedule and the S_t bucket (double counting); the "ageing schedule" was a two-factor crude-rate proxy rather than $\sum_a m_a P_a$; its U_0 prior differed from the central scenario's, so its higher end-2025 stock reflected a different baseline prior rather than independent vital evidence; its expert-coded inputs used subjective feature weights with no sensitivity mapping; the transfer from generic excess-mortality methods to Cuban reporting institutions was never validated out of sample; and the reconstructed population remained migration-identified regardless. Its code and artifacts remain in the repository [Pérez-Riverol, 2026] for anyone wishing to inspect the retired path.

Table 2: Scenario medians (end-2025), on the official end-2021 base. These reproduce Table 2 of the main paper; the supplement previously carried a superseded set on a retired second base.

Scenario	Main assumption	Loss (M)	%	Pop. (M)
Conservative	ONEI-anchored	1.97	17.8	9.14
Central (crisis-adjusted)	migration prior above ONEI + bounded D_{fac}	2.53	22.8	8.58
Upper stress	analog upper bound	2.79	25.1	8.32

5 Data pointers

Scripts and JSON artifacts for every model, retired ones included, live under `demographics/scripts/` and `demographics/artifacts/` in the companion repository [Pérez-Riverol, 2026]. Adversarial notes: `artifacts/model_e_adversarial_notes.md`.

References

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Juan Carlos Albizu-Campos Espiñeira. La caída de la esperanza de vida al nacer en cuba. de la crisis sanitaria a la humanitaria. Horizonte cubano, Cuba Capacity Building Project, Columbia Law School, June 2023. URL `https://horizontecubano.law.columbia.edu/news/la-caida-de-la-esperanza-de-vida-al-nacer-en-cuba-de-la-crisis-sanitaria-la-hum` Published 18 June 2023. Source of the 71.25-year 2021 estimate, and of 73.14 excluding COVID-19 and 73.90 for women. The 73.7 figure is the OFFICIAL/UNDP value this article argues against, not an alternative estimate by this author; an earlier version of this note said otherwise.

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